

Senior High School



Lesson Exemplar in General Science

Quarter 3

LE

2

Lesson Exemplar for General Science
Quarter 3, Unit 2

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LESSON EXEMPLAR

Learning Area	General Science	Grade Level	11
Semester	2 nd	Quarter	3

I. OBJECTIVES (*Identifying the Goals*)

Content Standard	Structures and functions are related at all levels of the organization of life;
Performance Standard	Create a presentation using chosen multimodal activities (e.g., forum, infographic, podcast, vlog, etc.) to effectively communicate key concepts about the effects of climate change on the local ecosystem in the learners' community.
Learning Competencies	Create a diagram to show the relationship between the structural components of a biological system and their functions;
II. REFERENCES and MATERIALS (<i>Selecting Resources and Material</i>)	Department of Education. (2021). <i>Earth and Life Science Quarter 2 – Module 2: Unifying Themes in the Study of Life</i> . Lingcol, R. (n.d.). <i>Unifying themes in the study of life</i> [Presentation]. SlideShare. https://www.slideshare.net/slideshow/unifying-themes-in-the-study-of-life/241817637 Studocu. (n.d.). <i>Lesson 1: Unifying Themes of Life – Earth & Life Science Q2</i> .

(*These shall be accomplished per topic*)

III. CONTENT (<i>Sequencing Content</i>)	Unifying Themes of Life Science
IV. OBJECTIVES (<i>Setting Clear Objectives and Analyzing the Tasks</i>)	a. Explain the structural components and functions of a selected biological system((cells, tissues, organs, organ systems). b. Analyze how each component contributes to the overall function of a biological system. c. Develop a diagram showing the relationship between the structural components of a biological system and their functions
V. PROCEDURES	ANNOTATION



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<i>(Selecting Strategies, Making Meaningful Content, Delivering Lesson and Assessing Learning)</i> This section focuses on selecting learner-centered, evidence-based instructional approaches such as problem-based learning, collaborative tasks, interdisciplinary integration, and technology-enhanced instruction. These strategies are intended to foster active engagement, critical thinking, and adaptability across diverse learning pathways. The chosen approaches and methodologies will be reflected through varied and relevant activities and assessments that emphasize real-world relevance and application, thereby enhancing learner engagement and comprehension. (Each part shall have 2-3 varied activities)		<i>*Instruction to teacher on how to facilitate the activities.</i> <i>*In the Annotation, explicitly explain how the IDF is applied in each part of the lesson</i>
A. Activating Prior Knowledge	Activity: From Cell to System: A Body of Connections” Instructions: 1. Warm-Up Prompt (5 mins): Ask learners: “Think of a time when your body responded to hunger. What parts of your body were involved—from the smallest unit to the whole system? - Let learners jot down quick thoughts individually or in pairs. 2. Group Sorting Challenge (10 mins): In small groups, learners receive a shuffled set of cards labeled with terms related to a biological system (e.g., epithelial cell, muscle tissue, stomach, digestive system). - Their task: Sort the cards into four categories: Cells → Tissues → Organs → Organ System - Then, draw arrows to show how each level connects to the next, adding functions beside each component.	Teacher management guide for the Activity: Preparation Checklist - Prepare card sets labeled with biological terms (cells, tissues, organs, organ systems) - Provide large sheets of paper or whiteboards and colored markers - Print warm-up prompts and sticky notes for peer feedback - Arrange classroom for small group collaboration and gallery walk Step-by-Step Facilitation 1. Warm-Up Prompt (5 mins) - Ask learners to reflect on a real-life experience (e.g., feeling hungry) and identify body parts involved.



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3. Gallery Walk & Peer Dialogue (10 mins):

Groups post their diagrams around the room. Learners rotate, view others' work, and leave sticky notes with questions or affirmations (e.g., "I didn't know that the stomach has three muscle layers!" or "How does this tissue help digestion?").

- Encourage journaling or pair-sharing to activate prior knowledge and personal connection.

2. Sorting & Mapping Challenge (10 mins)

- Divide learners into groups of 4–5.
- Distribute shuffled cards and materials.
- Guide learners to:
 - a. Sort terms into hierarchical levels:
Cells → Tissues → Organs → Organ System
 - b. Draw arrows and annotate functions
 - c. Use color coding or symbols to show relationships

3. Gallery Walk & Peer Dialogue (10 mins)

- a. Have groups post their diagrams around the room.
- b. Learners rotate, observe, and leave sticky notes with questions or affirmations.
- c. Encourage respectful dialogue and curiosity.

4. Whole-Class Reflection (5 mins)

- Facilitate discussion using guiding questions:
- a. What patterns did you notice?
 - b. How do small components contribute to system function?



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		c. What surprised you about another group's diagram? Differentiation Tips a. Provide visual aids or simplified card sets for learners who need support b. Allow verbal explanations for learners with writing challenges c. Offer extension: challenge advanced learners to compare two systems or add dysfunction scenarios
B. Instituting New Knowledge	Mini Lecture: "From Cells to Systems: Why Every Part Matters" Part 1: The Hierarchy of Life (2 minutes) Main points of the discussion: • Cells are the basic units of life—each with a specific job. Think of them as individual workers. • Tissues are groups of similar cells working together—like departments in a company. • Organs are made of multiple tissues that perform a focused task—like a machine built from different parts. • Organ Systems are networks of organs that collaborate to keep you alive—like a team completing a mission. ("This hierarchy isn't random—it's designed for efficiency, specialization, and resilience.")	Teacher management guide: Preparation Checklist • Review the mini lecture script and adapt examples to the biological system studied (e.g., digestive, circulatory) • Prepare a visual aid (e.g., diagram of the hierarchy: cells → tissues → organs → systems) • Ensure a quiet, focused environment for reflection and discussion • Prepare a whiteboard or digital board for key takeaways. Facilitation Steps 1. Transition from Activity (1 minute)



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	Part 2: Interdependence in Action (2 minutes) • Every level supports the next. A damaged cell can affect a tissue. A failing organ can disrupt an entire system. • Example: In the digestive system, epithelial cells absorb nutrients, muscle tissues move food, organs like the stomach and intestines coordinate digestion, and the system ensures energy reaches every part of your body. (“Just like in a community, when one part struggles, others feel the impact. That’s interdependence.”) Part 3: Why It Matters (1–2 minutes) • Understanding this structure helps us care for our bodies—through nutrition, rest, and awareness. • It also teaches us a deeper lesson: every part has value, and collaboration leads to function. (“Whether you’re a cell in a tissue or a person in a team, your role matters. Biology isn’t just science—it’s a mirror of how life works.”) Closing Reflection Prompt <i>“Take a moment to reflect: What part of the system do you relate to most today—and why? Are you absorbing, supporting, moving, or coordinating?”</i>	• Acknowledge learners’ effort during the “System Snapshots” activity. Say: <i>“Now that you’ve mapped out how the parts of a system connect, let’s take a moment to reflect on what that really means—and why it matters.”</i> Deliver the Mini Lecture (5 minutes) • Use the prepared script, adjusting tone and pacing for clarity and engagement. • Emphasize: 1. The hierarchy of biological organization 2. The interdependence of components 3. The real-life relevance and values connection 3. Facilitate Reflection Prompt (1–2 minutes) Ask: “Which part of the system do you relate to most today—and why?” • Allow learners to share verbally or write in journals. • Encourage connections to roles in teams, families, or communities.
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Values Integration:

- **Interdependence**
Highlight how each biological part supports the whole—mirroring teamwork and community
- **Stewardship**
Connect understanding of body systems to making informed, healthy choices
- **Self-awareness**
Use the reflection prompt to help learners see themselves as part of larger systems

Assessment & Feedback

Tool	Purpose
Oral responses or journal entries	Gauge conceptual understanding and personal insight
Exit slip (optional)	Ask: “What’s one thing you learned about systems today that applies to life outside biology?”
Observation	Note engagement, attentiveness, and connections made during discussion

Differentiation Tips



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		• Use visuals or analogies (e.g., orchestra, company, sports team) to support diverse learners • Allow written or verbal reflection responses • Invite learners to draw their own “system self-portrait” as an extension
C. Demonstrating Knowledge and Skills	Activity: “Inside-Out: Modeling a Living System Activity Overview Learners will work in small groups to create a 3D model or interactive diagram of a selected biological system, accompanied by a brief oral presentation and a function map that explains how each component contributes to the system’s overall function. I. Materials Needed • Recyclable materials (boxes, tubes, paper, string, etc.) or digital tools (e.g., Canva, Google Slides) • Manila paper or poster board • Markers, glue, scissors • Rubric for assessment • Optional: tablets or laptops for digital diagrams II. Procedures 1. System Selection & Research (15–20 mins) a. Each group selects a biological system (or is assigned one).	Teacher Management Guide for the Inside-Out: Modeling a Living System Activity Preparation Checklist • Assign or allow selection of biological systems per group • Prepare recyclable materials or digital tools for modeling • Print rubrics and peer feedback forms • Prepare visual aids or reference diagrams for support • Arrange classroom for group work and presentations • Facilitation Steps & Prompts 1. Introduction & Framing (10 mins) • Begin with a prompt: <i>“Think of your body as a team. What roles do different parts play to keep you alive and functioning?”</i>



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	b. Learners research and list its key components: cells, tissues, organ, and the overall system. c. They identify each part's function and how it supports the system. 2. Model or Diagram Creation (30–40 mins) a. Groups build a 3D model or interactive diagram showing: b. Structural hierarchy (cells → tissues → organs → system) c. Functional relationships (e.g., arrows, color coding, labels) d. Include a function map: a written or visual explanation of how each part contributes to the system's function. 3. Presentation & Peer Review (15–20 mins) a. Each group presents their model/diagram to the class. b. Peers ask questions and give feedback using a structured form (e.g., "I learned...", "I wonder...").	Connect to prior activity or mini lecture: <i>"Today, you'll demonstrate your understanding by building a model or diagram that shows how each part—from cells to systems—works together."</i> 2. System Selection & Research (15–20 mins) Guide learners to identify: a. Key components (cells, tissues, organs, system) b. Functions of each component c. How each part contributes to the whole Ask: <i>"What happens if one part fails? How does that affect the rest of the system?"</i> 3. Model/Diagram Creation (30–40 mins) ● Encourage creativity and clarity: Instruct learners to use color, arrows, labels, and structure to show how the parts connect and function." ● Circulate and ask: 1. "What does this tissue do?" 2. "How does this organ support the system's goal?"
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3. “Can someone unfamiliar with biology understand your diagram?”

4. Presentation & Peer Review (15–20 mins)

a. Each group presents their model and function map

b. Use peer feedback forms with prompts:

“I learned...”

“I wonder...”

“One suggestion I have is...”

Values Integration Tips

- **Interdependence.** Highlight how each part supports the whole—like roles in a community
- **Stewardship.** Connect understanding of body systems to personal health and care
- **Creativity.** Encourage innovative use of materials and visual storytelling
- **Collaboration.** Foster teamwork and respectful peer dialogue during presentations.

Assessment Tools

Tool		Purpose
Rubric		Evaluate accuracy, clarity, creativity, and presentation skills



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		Peer Feedback Form	Promote reflection and constructive critique
		Teacher Observation	Monitor group dynamics, engagement, and conceptual understanding
GENERALIZATION: The System Speaks: A Reflective Synthesis” Activity Description Learners will write a short reflective piece or create a concept map that synthesizes their learning by imagining the biological system as a speaking entity. This metaphorical approach encourages systems thinking, personal insight, and values integration. 1. Prompt Introduction (3 mins) Say: <i>You’ve explored the parts of a biological system—how they’re built, how they work, and how they connect. Now, imagine the system</i>		Differentiation Tips • Provide visual references or simplified templates for learners who need support • Allow verbal explanations or digital alternatives for learners with writing challenges • Offer extension: challenge advanced learners to compare two systems or simulate dysfunction scenarios	
		Facilitation Steps & Prompts for the Generalization Set the Tone (3 minutes): Say: <i>“You’ve explored how cells, tissues, organs, and systems work together. Now, let’s step back and listen—if your biological system could speak, what would it say about its parts, its purpose, and its power?”</i> Display the two task options: Option A: Write a short monologue from the system’s point of view	



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could speak. What would it say about its structure, its teamwork, and its purpose?”

2. Student Task (10–15 mins)

Choose one of the following formats:

Option A: Reflective Monologue. “Write a short monologue (5–7 sentences) from the point of view of the biological system. Let it explain its parts, how they work together, and why each one matters.”

Option B: Concept Map with a Message: “Create a concept map showing the hierarchy (cells → tissues → organs → system) and add a central message that captures what the system wants us to understand about its function and unity.”

Option B: Create a concept map with a central message

Guide the Task (10–15 minutes)

Encourage learners to choose the format that best expresses their understanding.

- Circulate and prompt with questions:
 - a. “What message does your system want to share?”
 - b. How does your diagram show the flow from cells to systems?”
 - c. “What happens when one part doesn’t function well?”

Optional Sharing & Reflection (5 minutes)

- Invite volunteers to read their monologues or explain their concept maps.
- Use prompts to deepen reflection
- What insight did you gain about how systems work?”
- How does this apply to how we care for our bodies—or even how we work in teams?”

Assessment Tools

Tool	Purpose
Reflective Monologue or Concept Map	Assess synthesis, systems thinking, and creativity



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		<table><tr><td>Teacher Observation</td><td>Monitor depth of insight and engagement</td></tr><tr><td>Optional Peer Feedback</td><td>Encourage appreciation and respectful critique of others' work</td></tr></table>	Teacher Observation	Monitor depth of insight and engagement	Optional Peer Feedback	Encourage appreciation and respectful critique of others' work
Teacher Observation	Monitor depth of insight and engagement					
Optional Peer Feedback	Encourage appreciation and respectful critique of others' work					
	EVALUATING LEARNING: System Check: Performance Task & Reflection” 1. Task Briefing (10 mins) Present the task: “<i>You’ll demonstrate your learning by creating a clear, accurate diagram of a biological system, explaining each part’s function, and reflecting on how the system works as a whole—and why that matters.</i>” 2. Student Work Time (40–50 mins) Learners complete the following: • Diagram: Cells → Tissues → Organs → System (with labels and arrows showing relationships) • Function Table: A chart listing each component and its role	Differentiation Tips • Provide sentence starters or concept map templates for learners who need scaffolding • Allow audio or video reflections for learners with writing challenges • Offer extension: ask learners to compare two systems or write from the perspective of a malfunctioning system FACILITATION STEPS AND PROMPTS FOR THE EVALUATING LEARNING Preparation Checklist • Print or distribute the performance task instructions and rubric • Prepare sample diagrams and function tables for reference • Set up a quiet, organized workspace for individual assessment • Prepare peer/self-assessment forms (if applicable)				



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	- Reflection Questions: “How do the components of this system depend on each other?” “What happens when one part fails?” - Why is it important to understand this system in real life?” 3. Peer Review or Self-Assessment (10–15 mins) Learners exchange work or use a rubric to assess their own performance. Prompts: “Is my diagram complete and accurate?” “Did I explain each function clearly?” “Did I show how the parts connect?” “Did I reflect on the system’s real-life relevance?”	- Ensure access to materials (paper, colored pens, digital tools if needed) Introduction & Framing (10 minutes) Set the tone: “<i>Today, you’ll demonstrate what you’ve learned about biological systems—not just by naming parts, but by showing how they work together and why they matter.</i>” Review the task components: - Labeled diagram - Function explanation table - Systems analysis - Personal reflection Student Work Time (40–50 minutes) Encourage learners to work independently and manage their time: 20 mins: Diagram and labeling 15 mins: Function table 10 mins: Reflection writing 5 mins: Review and finalize Circulate and prompt with guiding questions: “How are you showing the connection between structure and function?” “What message does your system diagram communicate?”
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		“How does your reflection connect biology to real life?”				
V. ASSESSMENT <i>(Assessing Learnings)</i>	Part 1: Knowledge & Understanding (Written Response) Instructions: Choose one biological system (e.g., circulatory, nervous, plant transport, immune). Answer the following: - List and describe at least five key components of the system. - Explain how these components work together to maintain balance and function. - Identify one feedback mechanism or regulatory process within the system.					
	Part 2: Systems Thinking (Diagram Construction) Instructions: Create a labeled diagram of your chosen biological system. Your diagram should: - Show the components and their functions - Use arrows or symbols to illustrate interrelationships - include brief annotations explaining flow, regulation, or collaboration					
	Suggested Rubrics					
	Criteria	Excellent (5)	Proficient (4)	Developing (3)	Emerging (2)	Beginning (1)
	Component Identification	All key components (cells, tissues, organs, system) are accurately identified and labeled	Most components are correctly identified and labeled	Some components are missing or mislabeled	Many components are missing or unclear	Few or no components are correctly identified
	Function Annotation	All components include clear, concise explanations of their functions	Most components include accurate function explanations	Some functions are vague or partially correct	Few components have function explanations	Functions are missing or mostly inaccurate
Interrelationships & Flow	Arrows/symbols clearly and accurately show logical	Most relationships are shown with	Some relationships are unclear or	Arrows/symbols are used but	No clear relationships or flow are shown	



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		relationships and flow between components	appropriate arrows/symbols	inconsistently shown	inaccurately or confusingly	
	Systems Thinking	Diagram shows deep understanding of how parts collaborate and regulate system function	Shows good understanding of component collaboration and regulation	Shows basic understanding; some connections are unclear	Limited understanding of system integration	Lacks evidence of systems thinking or integration
	Visual Organization & Clarity	Diagram is neat, well-organized, and easy to interpret; visual elements enhance understanding	Mostly neat and organized; minor issues with clarity	Some clutter or confusion; labels or layout need improvement	Difficult to follow due to poor organization or missing labels	Disorganized or incomplete; hard to interpret
Scoring Guide 21–25 points: Outstanding systems understanding and visual communication 16–20 points: Solid grasp with minor gaps or visual issues 11–15 points: Partial understanding; needs more clarity or accuracy 6–10 points: Limited understanding; major gaps in content or structure 1–5 points: Minimal effort or incomplete submission						
VI. REFLECTION <i>(Feedback and Continuous Improvement)</i>	<i>This section presents the key highlights and challenges encountered by both teachers and learners during the teaching-learning process throughout the unit. It also includes the adjustments made by the teacher to improve instruction.</i>					